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Urban traffic flow prediction techniques: A review

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ABSTRACT

In recent decades, the development of transport infrastructure has had a great development, although traffic problems continue to spread due to increase due to the increase in the population in urban areas that require the use of these means of transport. This has led to increased problems related to congestion control, which has a direct impact on citizens: air pollution, fuel consumption, violation of traffic rules, noise pollution, accidents and loss of time. In Latin America, the disorderly growth of cities increases distances and routes, likewise, there is an accelerated increase in the number of cars and motorcycles, which increases the problem. In this sense, intelligent transport systems are an alternative to improve the traffic environment, they incorporate the internet of things and intelligent algorithms, for the collection of data from multiple sources and information processing, respectively, in order to improve the efficiency of the transport flow. However, the processing and modeling of traffic data is challenging due to the complexity of road networks, the space–time dependencies between them, and heterogeneous traffic patterns. In this review study, (i) the smart techniques used for the analysis of mobility data in the prediction of traffic flow in urban areas are grouped, likewise, (ii) the results of implementing said techniques are shown, in addition, (iii) The procedures performed are described and analyzed to understand the benefits and limitations of these smart techniques. Given the above, (iv) the data sets used in the literature and available for use are shown, in addition, (v) the quantifiable results of precision of the various techniques were compared, highlighting advantages and limitations, which allows us to (vi) identify the related challenges and, from there, (vii) propose a general taxonomy in which the knowledge acquired in this traffic flow review converges from a computational approach.

1. Introduction

Currently, 55% of the world's people live in cities and it is estimated that this proportion will increase by 13% by 2050 [1]. This has led to increased problems in transport systems, including those related to congestion control, which has a direct impact on citizens: in terms of environmental pollution, accidents and infractions of traffic regulations, traffic, and increased travel times [2]. Despite the numerous measures taken to lessen the serious consequences of the growth of vehicles, the problem still exists, hampering the speed of urbanization, especially in some cities [2].

The US Department of Transportation defines traffic congestion as one of the most important threats to economic success [3]. Each year, congestion costs in the United States are around $1200 [4] en [5]. An effective forecast of traffic flow helps alleviate the spread of congestion, reducing pollutant emissions caused by the fuel consumption [6], positively impacting the sustainability of cities. Latin America is no stranger to the global situation, evidenced in the disorderly growth of peripheral areas of cities that increases distances and routes. Likewise, there is an accelerated growth in the number of cars and motorcycles that increase congestion, air pollution, traffic accidents, and increase the pressure to expand the road system, using public resources that are scarce [7]. According to the Tomtom Traffic Index, 2019, Bogotá, Lima and Mexico City are classified as cities with the highest vehicular traffic, with congestion levels that exceed 50% of overtime on each trip [8].

Transport systems continue to evolve in order to try to optimize them, in such a way that they can be framed within the concept of smart cities, being possible to demonstrate the following:

- Intelligent Transportation Systems (ITS) are emerging as an efficient alternative to improve the traffic environment [9]. ITS incorporates advanced technologies including electronic sensor

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technologies, including intelligent control and efficient data transmission, to supply information of interest to users of transportation systems.

- Recently, Internet of Things (IoT) has become an emerging research area within Information and Communication Technologies and, therefore, in ITS. IoT technology connects smart devices and sensors that can be accessed through the Internet. Internet-connected IoT objects send and receive data through various communication media such as Wi-Fi, RFID, WSN, NFC, Bluetooth, others [10]. IoT is flourishing due to its reliability, interoperability, and real-time interconnect capabilities for smart grid applications. In addition, the number of connected devices continues to increase, and with it, the amount of data collected by those devices is also increasing [11]. The progressive development of models for the analysis of traffic exploits the growing avalanche of data recorded by IoT devices, which, conflated with smart algorithms, present great potential for the applications that make up these systems [12].
- With the rapid growth of computational technologies, and the increasing volume of traffic data, much of the work is focused on developing intelligent models to predict traffic flows. Through these models, closely related to Artificial Intelligence, computer systems learn to perform tasks such as classification, grouping, predictions, pattern recognition, and others [11]. Machine learning is becoming increasingly popular for the traffic flow prediction task, for its high proficiency in extracting complex non-linear characteristics, overcoming statistical methods, and handling complex traffic forecasting problems [11]. Today, the emergence of deep learning techniques take advantage of the capabilities of artificial intelligence models to predict traffic more effectively [13].

This article reviews machine learning techniques for the analysis of mobility data for traffic prediction in urban areas, filtering literary articles no older than five years old, where the authors addressed the use of algorithms for predicting traffic flow in urban areas, taking into account that their results could be quantified. The reviewed articles were classified according to the technique used to forecast traffic flow and in each one of them a brief methodological description of their proposal, the data set used, as well as the results obtained through evaluation metrics is highlighted quantifiable. A comparison of the precision results obtained is made, and opportunities are detected to improve the current models in order to find more efficient precision results.

The content of the article is divided as follows: in Section 2, the existing models applied in the prediction of traffic flow are reviewed and categorized according to the type of algorithm. In Section 3.1 a general scheme for the prediction of traffic flow is proposed, based on the existing challenges in the literature. In Section 3.2 the metrics used by the authors to evaluate the prediction models are identified. In Section 3.3 the revised techniques are analyzed according to the prediction results, data set and evaluation metrics used. Section 3.4 presents an analysis of the opportunities detected in the review that allows us to generalize the challenges still present in the area of urban traffic prediction. Finally, the conclusions are shown in Section 4.

2. Literature review

The increasing volume of traffic flow has affected the sustainable development of cities. Various authors have explored a number of techniques to forecast traffic flow with high precision, taking into account different aspects and related characteristics. In general, three large categories of traffic flow prediction models can be found: (i) parametric techniques, (ii) machine learning techniques and (iii) deep learning techniques. In Fig. 1 we proposed a taxonomy of the techniques reviewed in the literature.

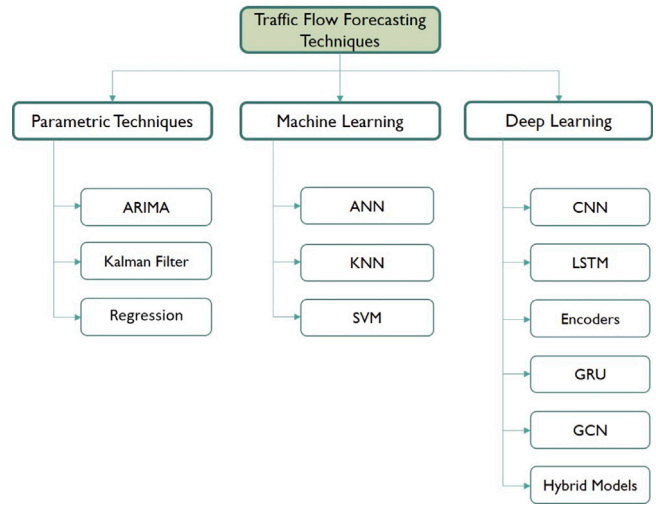

Fig. 1. Taxonomy proposed of techniques used in the prediction of urban traffic flow.

2.1. Parametric techniques

In the last decade, various strategies and methodologies have been proposed to address problems related to traffic flow forecasting. Kalman filters and the integrated auto-regressive moving average (ARIMA) are the most studied statistical techniques for forecasting time series data. However, they must be trained on a large data set to achieve better precision results. Emami et al. [14], taking into account that the recent development of sensor technology connected to the Internet facilitates the possibility of developing forecasting models more effectively, they propose a method based on adaptive Kalman filters to diagnose the future flow of traffic, merging the data supplied by connected vehicles and Bluetooth devices [14]. The model uses a test bed of a scheme of more than 100 connected vehicles, around a 6 km^2 urban catchment area in the Australian city. The results show that as vehicle volume increases, more data is generated and, therefore, the Kalman filter performs flow forecasting more effectively. The proposed method works best with a significant margin, close to 11% [14].

Statistical methods, compared to other current methods, have shown less precision [15], and although these techniques can extract time features from time series, they present difficulties in representing spatial relationships in traffic flow forecasting tasks [16]. Several authors have proposed a combination of these methods with artificial intelligence models to solve the above problems. Hou et al. [17], on the topic of traffic flow uncertainty, caused by dynamic change in traffic structure, reports that the limitations of prediction approaches can be partially overcome by using the use of combined models, and from this they propose an adaptive hybrid model to forecast vehicle behavior in the short term. They combine the ARIMA method and the Non-linear Wavelet Neural Network method to predict traffic flow. When evaluating the proposed model, the results show greater efficiency in predicting traffic flow than the two individual models, whether under stable or fluctuating conditions. However, the results only highlight the hybrid model against the individual models, but do not take into account other models that have demonstrated greater efficiency than ARIMA [18], such as machine learning or deep learning models. Likewise, they do not consider other characteristics of the environment that influence the dynamics of traffic flow, such as weather conditions or events that attract crowds.

Tempelmeier et al., examine the literature and find reduced studies of the influence of events on the understanding of mobility needs in the city. Therefore, they propose a supervised model to extract spatial and temporal dependencies in traffic flow data taking into account the incidence of events. To enable accurate predictions, they use a

variety of event features, mobility behavior, and urban infrastructure. The regression model has the ability to predict the spatial dimension of the effect of events, using k-Nearest Neighbor algorithms (KNN) and Ridge regression algorithms and support vectors (SVR). They apply the methodology in approximately 195 million historical data on the behavior of mobility and information on the occurrence of events in Hannover (Germany), in a period of four months in which 150 important urban events occurred. The results show a mean absolute error (MAE) with a reduction of up to 27% [19].

Shen et al., propose a traffic flow forecast architecture for multiple intersections, using a vector regression of relevance (RVR) model based on chaotic optimization. They conducted an experimental analysis using traffic flow data in the cities of Minnesota and Rochester. Shen et al., using evaluation metrics such as mean square error (RMSE), MAE and percentage error (MAPE) showed that its method has better precision ability. The multi-intersection model also surpasses the ordinary model in five of the six sections studied, with RMSE values ranging from 5.98 to 29.62, compared to 7.56 to 35.56 [20]. Previous studies use SVM and Relevance Vector Machine, they made reference to the fact that the execution time used by the parameter optimization algorithm represents 80% to 95% of the total training time [21], and its efficiency negatively impacts the performance of the algorithm in general [21].

Bratsas et al., compares four methods (support vector regression, multiple linear regression, neural networks and random forests) with the intention of forecasting the speed of vehicular traffic, in the Greek city of Thessaloniki, for an interval 15 min prediction. The results show that when applying the MAE metric, the SVR-based model presents an error of 6.25% and is more effective than the other models at approximately 0.2 km/h [22]. The SVR and neural network models performed better when evaluated by RMSE. The sudden changes in speed influenced the effectiveness of the linear regression model, however, all the methods present a delay in the execution of the calculations. Furthermore, these are also considered black box methods, which means that the results are not easy to analyze or draw conclusions [23].

Tao et al., points out that the ad-hoc vehicular network, known as VANET, can contribute positively to the operational management of traffic flow. However, the large volume of traffic and, therefore, vehicles in the traffic network, would cause instability in its topology in terms of online information processing. Based on previous research, they found that travel delay is a major problem that could affect forecasting results. This is how Tao et al., propose a hybrid autoregressive algorithm, based on vehicle travel delays to improve forecast results. According to Tao et al., in previous spatiotemporal models, this factor is not considered. The proposed scheme uses matrices to represent the space–time delay of the trip in a set of traffic data in England, with intervals of 15 min, taking into account that the days on which the data were collected were every Wednesday in May 2015. Seven different routes, with different flow volumes, were chosen to compose the study data set. The result shows the reliability and ability of the proposed algorithm to deliver data with better precision to support the VANET network. The proposed model showed the lowest MSE and MAPE metrics, but the highest R2 for all three locations. The best result of the model is presented with MSE and MAPE of 484.51 and 4.92%, compared to 6325.88 and 12.35 of the ARIMA method [24]. However, the selection of data collected does not adequately describe a representation of the traffic flow in the week or the month.

2.2. Techniques based on machine learning

Methods based on machine learning can more accurately predict traffic flow, saving labor and material resources [16]. Various investigations exploit machine learning in predicting urban mobility data, due to its non-linear and stochastic behavior [6], to extract patterns and create adequate forecasting models [25]. In this section we highlight some authors who use machine learning models to solve problems related to the heterogeneity of data that can come from different sources, types and characteristics, quantification of space–time dependencies, strong non-linear characteristics of traffic flow, and the various associated dynamic and static influence factors.

In that order, Ata, et al., have proposed a machine learning model that uses Support Vector Machines (SVM), to analyze data from multiple sensors, in order to forecast traffic congestion, but not before preprocess data to deal with redundant or missing information. The algorithm to analyze the traffic congestion proposed, during the training, showed levels of sensitivity, specificity, precision and failure of 98%, 99.5%, 98.7% and 1.3%, respectively, for Validation obtained values of 97.56%, 98.37%, 97.9% and 2.1%, respectively [5]. However, the effectiveness of the algorithm can be affected by delays in data taken from previous processes. Also, the continuous acquisition of information to the model can affect the complexity of the data [5].

Cheng et al., consider the spatial characteristics of the road network as a function of adaptive spatial neighbor nodes, time intervals, and space–time equivalence values [26]. They propose an adaptive KNN model evaluated on two sets of vehicle speed data collected on highways in California, USA, and on urban roads in Beijing, China. The result proves to be more efficient compared to other techniques, with MAPE values lower than 13.5%, RMSE lower than 12% and MAE lower than 9%. The level of precision of the prediction varies according to the level of the spatial scale [27]. By introducing clustering methods to portion time periods appropriately and capture non-stationary temporal dependencies, the proposed model could improve its accuracy [26]. Likewise, the proposed algorithm shows slightly different behaviors in all validated time slots. However, to appease the computational cost for online executions, a parallel adaptive KNN algorithm is suggested in an existing parallel computing framework [26].

Cai et al., Taking into account the existing spatio-temporal correlation in the traffic network, propose an improved KNN model that adopts the use of a novel dynamic distance measure that uses physical and data features to replace the physical distance and makes representations in state matrices to model the traffic situation. The proposed scheme is evaluated on the traffic data set of the Liuliquiao district, in Beijing, and compared with parametric techniques and other intelligent algorithms. The proposed model shows greater precision with MAPE values 2.96% higher than the original KNN model. Thus, the best result of the proposed algorithm obtains an RMSE of 0.05 and a MAPE of 6.99%. Although the proposed research contemplates the spatio-temporal correlation in the study of the traffic state, Liu et al. [28] considers that the models based on the spatio-temporal correlation are not suitable in road networks urban, due to the linear relationship of the correlation between the two variables.

Tang et al., Indicate that external factors cause interference in the raw traffic flow data and introduce noise that can cause a decrease in prediction performance. Therefore, the authors propose a forecasting method that combines SVM and denoising schemes to improve accuracy [29]. The authors evaluated the model with different denoising algorithms including Moving Average, Butterworth filter, Empirical mode decomposition, and Wavelet, on traffic volume information collected from three highway loop detectors in the city of Minneapolis. The experiment reveals that the proposed algorithm presents the best precision results, with respect to other methods, through the MAE, RMSE and MAPE metrics of 2.59, 3.34 and 15.28 respectively for a time scale of 2 min, and 8.03,10.63 and 6.26 respectively for a 10 min time scale [29]. The results show better performance in the model combined with the denoising algorithm and provide indications for the proper selection of denoising model. However, Zhang et al., [30] further suggests merging multiple sources of mobility data, such as GPS information and meteorological data, to improve prediction accuracy.

Liu et al., Recognize the strong non-linear characteristics of urban traffic flow, the multiple dynamic and static influence factors involved, and the increasing challenge of predicting traffic flow in the urban environment [31]. Therefore, through the study called Chaotic Particle Swarm Optimization — SVM (CPSO/SSVM), it collects data from several sources: traffic flow data, data on weather conditions and data

from points of interest from different sources in 717 intersections in Guizhou City, China, removes all invalid records and uses descriptive statistical analysis to estimate intersections of interest with the highest average flow rate. The normalization algorithm is applied to unify the metric in all the variables of the model. Using a second-order kernel function in the SSVM algorithm, adaptive optimization problems of the CPSO algorithm are solved to generate prediction results. The relative error of the proposed algorithm is within 5% and the precision and stability of the model satisfy the forecasting needs of real traffic flow [31]. However, although the SSVM algorithm works well for less fluctuating data, it has little generalizability when it comes to fluctuating traffic with high intensity [31].

Lin et al., According to [32], who mentions that relevance vector machines have excellent generalization and extremely short test speed, they research a traffic conditions forecast approach based on Relevance Vector Machines to Optimize Genetic Algorithm Kernel and Particle Swarm Based on Optimized Stack Parallelization [21]. First, they build kernel functions by combining common functions. They then propose to optimize the model parameters for these functions using the genetic algorithm and particle swarm optimization. The model is evaluated on a data set provided by Whitemud Drive in Canada. The results are more accurate than other forecasting models, and the parallelization technology significantly reduces the time consumed by the parameter optimization algorithm. The proposed model is shown to be superior to other comparison models, with RMSE error values of 19.81, 12.69 and 6.41 for the three selected traffic sections [21]. However, for the optimization of the parameters, the convergence speed is very fast in the first training cycles, in most of the models, tending to reduce its speed in the subsequent iterations [21].

Liu et al., They acknowledge that traffic flow data in most of the road sections are not available, owing to the low volume of plate recognition devices and the changing flow circulation between the various sections, which hinders effective learning [33]. To solve this, they propose to merge plate recognition information with the help of cell localization, using machine learning algorithms to capture the traffic flow volumes of the transport network. The model takes into account the spatio-temporal characteristic of the data extracted from the city of Nanjing, China, with 24 plate recognition sites that generate approximately 130,000 daily records. In addition, to improve prediction, they incorporate multigrain functions and a sliding window to produce more samples. The results show a MAPE of 29.5%, 16.3% and 13.0% for time windows of 10, 30 and 60 min, respectively [33]. They argue that the precision of the model is quite good, considering that the flow is estimated in links without training samples. This work was demonstrated in rectangular subnets of a metropolitan road network, which have a relatively simple topology. In addition, when collecting cell location data, the estimate of traffic flow can be influenced by data from the neighboring population in the sections of the roads [33].

2.3. Deep learning techniques

Deep learning techniques are an extension of machine learning, composed of a large volume of processing layers capable of learning characteristics with a high level of abstraction. In recent years, it has been chosen to capture mobility parameters in cities through a variety of sensor devices, which have provided a large amount of data of various types and characteristics. At the same time, the recent deep learning approach purvey insight into high-dimensional data mining [6]. These techniques have notably improved the prediction results in vehicular traffic flow, with high capacities for the extraction of space–time dependencies in heterogeneous data sets from multiple sources. Among the deep learning techniques reviewed in this work within the framework of vehicular flow forecasting algorithms are: Convolutional Neural Networks (CNN), Short–Long-Term Memory Networks (LSTM), Encoders, Gated Recurrent Units (GRU), Graphical Convolutional Networks (GCN) and hybrid models that integrate two or more techniques to increase performance. This section will review the research carried out by different authors, using these techniques.

2.3.1. CNN technique

Driven by the flood of traffic data over the last decade and given that big traffic flow data still presents challenges as it contains abundant spatial and temporal characteristics, Wu et al., [34], proposes a model based on the CNN approach to extract the spatial features and GRU to know the temporal features of the traffic flow data in the data set *Caltrans Performance California Measurement System* (PeMS). To represent the multi-periodicity of the traffic flow, they entered into the prediction model, historical flow data, both recent and from the previous day, up to a week. Likewise, the velocity data was used to train an "attention" model to determine how strong the correlation of the historical and future flow data is. Using evaluation metrics such as MAE, RMSE and MRE, the proposed model showed an improved average performance compared to other methods, achieving MAE up to 19.1264 and MRE of 0.0700 [34]. However, error rates were found to increase at other prediction points. Likewise, the execution of the algorithm was demonstrated in rectangular subnets of a metropolitan road network, which have a relatively simple topology.

Fouladgar et al., Find that most of the existing traffic flow prediction systems employ a central processing architecture, which could have drawbacks of scalability and provision of real-time information to the system [25], propose a decentralized method based on a CNN in which each node accurately predicts its congestion state, based on the current state of neighboring stations. They emphasize that since it does not require historical data from the deployment site, the proposed method is suitable for scaling new stations. The results indicate, on average, a mean square error (RMSE) composed of values from 0.002 to 0.018 around the 30 points of the network. The normalized average speeds, foreseen for the proposed model, are of the order of 80% in night hours and between 70% and 80% in peak hours during the 30 days. It is observed that the error is proportional to the increase in congestion [25]. The proposed method attempts to forecast the traffic conditions in all sections of the network structure. However, it may happen that the forecast in a particular area is not executed properly, due to the large size of the deep learning architecture being modeled, which could make it difficult to model individual events [35].

The prediction of data sets with spatio-temporal dependencies is of great importance for the analysis of information ranging from urban planning, traffic control to public safety [36], [37]. Zhang et al., Based on historical trajectory, weather, and event data, introduces deep residual learning. This algorithm allows CNNs to have 100-layer super deep structure, even up to 1000+ layers [37]. The proposal uses the data sets of TaxiBJ (Beijing City Taxi GPS) and BikeNYC (New York Bicycle System Path Data). First, the input and output stream is converted to a 2-channel image. Then, the temporal dependency is made up of three sections, which express the recent time, the recent history and the distant history, and which represent three temporal characteristics: proximity, period and trend. These properties are modeled separately, in the same way through a CNN, followed by a sequence of residual units. This scheme extracts the spatial characteristics between neighboring and distant nodes. In addition, they manually extract characteristics from external data sets, such as event and rainfall information, to incorporate them into a fully connected neural network. The best precision results show that the RMSEs in both TaxiBJ and BikeNYC are 16.69 and 6.33, surpassing techniques such as ARIMA and the deep neural network proposed by [37]. Although these methods use historical timestamp traffic images for prediction, the time sequence dependency [38] is not explicitly modeled.

Lin et al., propose a CNN model that simulates long-range spatial dependence, considers the effect of location attributes and employs a fusion mechanism, to capture long-range spatial dependence of crowd flows in the city. They consider proximity, period, point of interest distributions and time information as flow input. The entrances and exits in each region are counted every hour or every half hour. The model is evaluated on two data sets: MobileBJ: the most popular social network provider in China, which records the locations of users whenever they

request the location service in the application; and BikeNYC: containing information on the duration of the trip and the location of departure and arrival stations. The proposed method yields an RMSE precision error of 36.29% and 5.984% in the data provided by MobileBJ and BikeNYC, respectively. It is observed that adding information on time and points of interest reduces the error of the model [36]. However, these methods mainly focus on modeling long-term characteristics for dense urban traffic situations. Likewise, in comparison with regions that have a less developed transport scheme, central cities should take into account neighborhood information [39]. The size of the convolution kernel that models the neighborhood dependency range is generally fixed and empirically established [39].

Based on the above, and the fact that current studies often use the residual network to aggregate the features in contiguous layers, without merging the low-level and high-level features [39], Li et al., propose a general network that incorporates a ST-NASNet (Neural Search Network for Spatial–Temporal Architecture) module that is used to learn the spatial–temporal network configuration, in different scenarios. Selected training and validation data from CrowdBJ (mobile base station indicating the flow of crowds in Beijing), TaxiJN (flow of taxis traveling through Jinan), TaxiGY (flow of taxis in Guiyang, via taxi GPS) and TaxiBJ show that the best precision The result of the proposed method occurs with a MAPE of 28.43% in the CrowDBJ data set. The authors also point out that the method has improved by 1.3%, 1.1%, 0.6% in the first three data sets, respectively, due to the decision-making capacity in the automatic fusion of the outputs of the preceding layers [39]. However, the proposed method does not take into account other factors that could affect prediction performance, such as rainfall, street classification, and spatial correspondence between nodes.

In recent years, local administrations are turning their sights to the problem of traffic forecasting. The most adopted measures are the use of predictive planning tools that seek to reduce the levels of accidents and polluting gases [40]. Impedovo et al., Propose the use of an adjusted version of the Google Deepmind Wavenet Architecture [41], through convolutional layers in which they use 12 filters, whose depth depends on the delay of the sliding window. The model evaluation is carried out on two data sets PeMS and TRAP-2017 (*Traffic Mining Applied to Police Activities*). The best prediction results are performed in PeMS, with MAE and RMSE of 6229 and 8668, with a slight improvement compared to LSTM with 6274 and 8734 respectively [40]. However, the training time is higher compared to other networks: 5293 in the proposed TrafficWave approach, versus 820 in LSTM.

2.3.2. LSTM technique

Lu, et al., find that current traffic forecasting methods have limitations in learning the high-dimensional temporal characteristics of traffic signs. Therefore, they combine a multicast convolutional block with a stacked LSTM block to address the spatial dependencies of traffic data with high-dimensional temporal characteristics [42]. The evaluation of the proposed method using the MAE, RMSE and MAPE metrics also exceeds the latest generation methods such as DCRNN [43] and GWaveNet [44], in 15, 30 and 60 min intervals [42]. The proposed model is validated on the public data set provided by METR_LA, collected by loop detectors on the Los Angeles County highway network highway. The MAPE value obtained was 6.82%, 8.35% and 9.89% for time windows of 15, 30 and 60 min respectively, and in PeMS of 2.68%, 3.59% and 4.88% for the same time horizons. However, while the precision results are encouraging, the authors emphasize that future studies consider economic and social factors for a more accurate forecast of traffic flow.

Wang et al. Note that both typical parametric and non-parametric traffic prediction techniques miss various factors that directly influence traffic flow, such as precipitation. They point out that the surface structure of recent architectures present difficulties in capturing the deep correlations of the mobility data in the vehicular network. Therefore, they propose a bi-directional memory LSTM scheme to capture the deep characteristics of the traffic flow, make the most of the traffic flow data over time and introduce climatological data. To check the accuracy of the proposed technique, the data sets of PeMS and KDD Cup 2017 are used, and comparison models such as ARIMA, SVR, LSTM. In the experimental results, with a horizon of 30, 40 and 50 min, the proposed method exceeds the comparison algorithms. It can be seen that for the PeMS data, with an interval of 30 min, the proposed method obtains a precision in MAPE as in RMSE with values of 3.23 and 110.43 respectively. When evaluated without precipitation data, it reaches 3.83 and 124.31; LSTM achieves 4.26 and 135.32 MAPE and RMSE respectively. The authors demonstrate that additional information from weather data improves the performance of the model [45]. A critical and apparent disadvantage of *seq2seq* models (like LSTM) is that they compress all information into a fixed-length vector, causing the inability to remember long tweets [46].

Wang et al., Considers that there is still an open problem about the exploitation of spatio-temporal characteristics of the traffic flow to improve its prediction and interpretability of the model. Therefore, they consider a route-based deep learning model to improve the prediction of traffic speed. Specifically, they identify critical routes in the city. Each critical path is then modeled using a bidirectional LSTM neural network. The model introduces multiple Bi-LSTM layers to add temporal characteristics to the network. In the traffic prediction stage, the captured spatio-temporal characteristics of the route are incorporated into a completely connected layer, for better speed prediction while maintaining strong interpretability. The results show that the critical path-based deep learning algorithm performed better. The evaluation results compared with ANN, KNN, CNN and LSTM show a reduction of the proposed model between 9.92% and 35.40%. In the short term, the proposed method has an average precision of 86%, and in the long term, it showed a precision of around 70% [47]. LSTM is shown to have a higher prediction error compared to the Bi-LSTM model based on critical paths and CNN. RNN models based on LSTM or CNN perform efficiently to process chronological data series, however they have difficulty capturing the spatial characteristics between detectors within the data set [9].

Yuhan Jia et al., Consider the impact of rain to predict traffic flow, applying deep learning methods, since in many investigations, they have not taken these factors into account, knowing that studies carried out in Beijing, show that the rainy climate notably influences mobility [48]. To achieve this, they propose two deep learning methods based on Deep Belief Network and LSTM to know the effective characteristics of traffic flow and rainfall data. DBN is used on the basis of the input layer, to obtain characteristics of interest quickly and without supervision, with each DBN training separately using raw data. For the output layer, tagged data is used to adjust the learned characteristics through a fine-tuning algorithm. The average prediction results for 10 min and 30 min were better compared to classical techniques such as ARIMA and the Back Propagation neural network. MAPE for 10 min in R-DBN was 13.14% and in R-LSTM it was 11.69%. The R-LSTM algorithm has shown a greater difference in error when incorporating rainfall data compared to data without taking into account rainfall (MAPE of 12.84% for 10 min of prediction) [18]. Accuracy was seen to improve when rain data was incorporated into its forecast algorithm, but its forecast horizon was a maximum of 30 min. The horizons of both algorithms are limited to recent observations of traffic conditions as inputs; Unusual or non-recurring traffic data, such as events or incidents that cannot be inferred from historical observations, are not taken into account.

Majumdar et al., Considering that traffic congestion in cities has had an unfavorable impact on environmental sustainability as a consequence of carbon dioxide emissions, and taking into account the enormous data set provided by the boom in devices IoT, smart and sustainable, transportation solutions can be provided within cities. However, the nature of traffic shaping is highly dependent on a number of interrelated parameters with complex and non-linear behavior,

which makes information processing an interesting challenge [12]. To solve this, they turn to the LSTM architecture to forecast the spread of congestion on a road network. The model collects speed data from sensing devices in two zones and predicts the spread of congestion every five minutes. The experimental results show that the proposed approach delivers acceptable precision performances of the propagation of congestion in short periods of time. It is considered that the precision of the proposed method ranges between 84% and 95% in a time window of five minutes, values close to the precision results of other models that use a similar LSTM scheme and present yields that range between 95% and 97% in two minute time windows [12]. Although LSTM simplifies the structure of the network with satisfactory results, its accuracy depends mainly on the design of the vehicular transport road network. Similarly, when applied in long-term and long-pass traffic flow prediction, the mean absolute error of them increases, which means that the overall RNN efficiency, LSTM will be slightly lost [49] .

Kouziokas, proposes a deep learning Bi-LSTM network layer model, to forecast traffic flow and contribute to management and security strategies traffic, taking into account environmental factors. The proposed scheme consists of two Bi-LSTM layers each composed of 250 hidden units. To deal with overfitting issues, layers of abandonment are built into the network. The data set used to feed the model was extracted from the traffic information system *PeMS* of the state of California, for a time horizon of 5 min. The collected climatological data for rainfall (inches/h) and visibility (miles) were extracted from the Automated Surface Observation System of the National Environmental Information Centers. The experimental results show that the proposed bidirectional Bi-LSTM model gives the best forecasting performance compared to other intelligent techniques evaluated in the same work. The RMSE metric evaluated in the proposed model is 6.5278 [50]. However, for the optimal configuration of LSTM they consider a pattern of trial and error and the interrelationship between the input and output information cannot be explicitly described. Therefore, interpretability is confusing [51].

2.3.3. Hybrid models

The growing development of intelligent computational tools, in conjunction with the Internet of Things, which has made it possible to have a large content of information in real time, have facilitated the prediction of traffic flow with greater precision, saving a large amount of labor. However, a large number of current models are generally implemented in road networks with relatively simple topologies [15]. Sun et al., Propose a CNN based on a bidirectional space–time network that uses traffic speed and meteorological information through data supplied by electrical maps. The convolutional network captures the spatial dependence of the traffic topology to compose the time series input data set, and an LSTM algorithm performs the data set training. By evaluating the proposed model against other machine learning techniques such as LSTM and CNN-LSTM, the authors verify its spatial and temporal extensibility, with promising results. The model was evaluated on a set of road traffic speed and weather data from Urumqi, China, over five months, with time windows of 1 min. The results of the experiment evaluated with meteorological data exceeds the scenario without meteorological information and the prescribed mean error is less than 5% [15]. The proposed model highlights the use of data from external sources with encouraging results as the basis for future studies. Therefore, with abundant data from multiple sources, the performance of the forecasting scheme can be remarkably contemplated [15].

In the last decades, the terms of prediction of traffic flow and traffic congestion have been considered in multiple investigations. For the traffic flow prediction problem, the key point is the treatment of space–time dependency. Taking into account that neighboring nodes tend to maintain a high connection, the local spatial characteristics, which represent the context of the flow through time, influence the progress of the trend. [52]. Du et al., proposes a multilayer hybrid model to improve the forecast of vehicular flow. CNN is used to characterize the spatial dependencies corresponding to the local projection, and an LSTM network models the timing and regularity characteristics of the traffic data. The data set used in the experiment is extracted from PeMS. The time period of the particular dataset used in this document is 02/01/2013 to 08/29/2013 (Station 4, District 5 min). The dataset is buffered by aggregating flow data without processing with 5 min time windows without overlapping. The experimental data set has 69,020 data points. 80% is used for training and the remaining 20% is tested on the model. An RMSE precision error of 0.028 [52] is observed in the results of the experiment. The authors note that the model improves accuracy. The experiment is compared to other algorithms including: ARIMA, RBF, Linear Kernel SVR, RNN, GRU, and LSTM, where it demonstrates better precision results.

Duan et al., combines a CNN to extract spatial characteristics, with LSTM to extract temporal dependencies, in the traffic flow forecast, using GPS taxi tracking. They propose a simple implementation version of reinforcement learning to improve accuracy when a network is deeper. The proposed technique has been shown to help improve prediction and reduce execution time. The results indicate a precision error measured in RMSE of 11.1519, compared to 14.1475 of the traditional CNN-LSTM approach cite duan2018improved. However, the model only took into account traffic data from taxis and does not consider other types of data that impact mobility and traffic flow as information from other vehicles such as trucks and buses, as well as information on weather conditions taking into account that the weather has a significant influence on the characteristics of the traffic flow [18].

Vijayalakshmi et al., agree on the need to use a model based on deep learning to process large volumes of data, and propose a CNN-LSTM algorithm. The scheme captures the spatial characteristics of the mobility data, to enhance the performance of the model. An attention-based architecture focuses on supporting the detection of short-term flow characteristics and considers the dependence between speed and flow for better prediction. Then, CNN and LSTM capture the space–time dependencies of the traffic data. Subsequently, the extracted features are merged to run the prediction. For the experiment, flow and velocity data from the PeMS traffic information system were used. The results highlight that the model has high precision in predicting short-term traffic in both congested and off-peak hours. The precision values obtained in terms of RMSE for 5, 15 and 30 min of interval are close to 2.2, 3.2 and 3.3, improving the results of other deep learning techniques [53]. To improve the model, factors such as weather, events and incidents such as accidents and information on the state of the road can be reviewed. The lack of data in real-world systems makes the prediction problem even more difficult; so it is important to properly plan a strategy to manage the data gap.

Yao et al., describes that when applying CNN in a graph that represents the traffic demand in an entire city, no encouraging results are obtained. In addition, he describes that forecast performance weakens when areas with weak correlations are taken into account to forecast particular areas. Taking into account the above, Yao et al., [38] presents a CNN-based model that only involves spatially neighboring areas, supported by observations of real data that suggest a strong correlation in neighboring regions. A region graph is used to extract semantic information, its edge representing the similarity of demand patterns between regions. An LSTM network captures spatial features and relates them to context features at the appropriate time. Then, a final layer that collects the CNN and LSTM characteristics is used to execute the respective forecast. The proposal is evaluated using real-world taxi demand data. The dataset contains around 300,000 daily taxi requests, provided by the DiDi taxi request platform in the Chinese city of Guangzhou, over a period of 2 months. The MAPE and RMSE evaluation metrics were 0.1616 and 9.642 respectively. In addition, during the weekends, the method also proved to be superior, obtaining an increase in the relative error of 4.04% [38]. However, the semantic information is implicitly modeled in this document. Furthermore, when

modeling temporal correlation, each neighborhood is processed independently [54]. It maintains that global and contextual information is also important. Although space–time correlation is taken into account, they are processed separately. Furthermore, input is limited to standard 2D or 3D grid data [54].

Yao et al., note that dependency based on daily and weekly periodicity of traffic data can be useful for prediction [55]. Although previous studies [37] considered periodicity, they did not consider sequential dependence and temporal change in periodicity. To address this challenge, Yao et al., propose a dynamic deep learning network architecture, which captures spatio-temporal characteristics through local LSTM and CNN algorithms. The local convolutional network captures the dynamic spatial characteristics of the traffic points. In addition, an attention mechanism changes periodically to extract the long-term periodic characteristics and the temporal change of the time series. The model is evaluated in the New York City public taxi (NYC-Taxi) and bike-sharing (NYC-Bike) data sets [55]. The evaluation results outperform other methods, both conventional and deep learning. The evaluation metrics for MAPE and RMSE are 16.25% and 19.05% at NYC-Taxi and 20.87% and 8.15% at NYC-Bike, respectively. However, CNN-based applications have limitations to handle the structure of the traffic network graph, because the topological traffic network has a structure like a graph [42,56].

2.3.4. Enconders technique

One of the relevant aspects that influence urban traffic flow is the way the flow changes in relation to neighboring nodes or points, due to the urban traffic network topology itself [57]. Wei, et al., based on the promising LSTM model for time series data due to its ability to learn long-term dependencies, proposes an algorithm that integrates autoencoder and LSTM, where the first subtracts the characteristics of neighboring points and the second makes the respective forecast of the acquired characteristics and the historical traffic flow. The test results indicate that the proposed algorithm achieves a better precision performance compared to other techniques such as SVM or CNN, reducing the error by 0.01, applying MRE [57]. However, this work only contemplates space–time series of a simple nature.

Chen et al. take into account many factors that can cause a traffic accident, such as driver behavior, weather, and road conditions. In addition, commercial and entertainment areas are always at higher risk of traffic accidents in most cases, which may be related to land use, or ultimately, to human activities. The reason is that these areas always have higher human density and higher population flow [58]. Chen et al., collect large amounts of heterogeneous data to understand how urban mobility can affect the risk of traffic accidents. Initially, they pre-process the traffic and mobility accident data. Then, they build a deep learning architecture based on the model called *Stack denoise Autoencoder*, to learn the representation of hierarchical characteristics of human mobility. Finally, with real-time mobility data entry, the model simulates the risk of large-scale traffic accidents and highlights high-risk regions. The model collects nearly 300,000 traffic accident records throughout Japan and GPS records from approximately 1.6 million anonymous users at the same location. The results indicate that the proposed model works better and the prediction error is lower compared to other competing methods such as Decision Tree, Logistic Regression and SVM. The model presents an MAE of 0.96, MRE of 0.39 and RMSE of 1.00. However, the complexity of the traffic accident data is considered a limitation. Human mobility data is not sufficient to build a satisfactory risk prediction model, as other variables may be included to improve performance, such as land use and POI data [58]. Although the proposed algorithm shows to obtain an encouraging prediction performance, it does not consider long-term memory and does not take into account the topological characteristics of the nodes in the road network, which could enhance the prediction results [59].

Today, around 40 percent of traffic jams in populated centers is due to the time to find parking, and this substantially impairs mobility [60]. Based on the above, Shaham and Siddiqui train a neural network algorithm to find free parking space. The model uses automatic logical and logistic encoders, in order to improve its performance, through a substantial reduction in memory use and computational training times. Initially, the data was pre-processed to eliminate anomalies. The training results are 94.37% accurate and the error rate is 5.66%. During testing and validation, the precision was 91.25% with an error rate of 8.75% [61]. However, the method is not compared to other methods to verify its effectiveness. Also, the database used has a small number of samples.

Zhang et al., performs an image analysis to create a traffic congestion data set, relying on public images of traffic congestion maps provided by the Washington State Department of Transportation [62] . Likewise, they present an algorithm that uses autoencoder with symmetric layers to achieve the learning of the temporal dependencies of the data to predict traffic congestion. In the experiment, the proposed method manages to effectively learn the temporal dependencies of the transport network for the forecast of vehicular congestion, with a weighted MAE and MSE prediction error of 0.0106 and 0.0579, surpassing comparison techniques such as Conv-LSTM [63] and SRCN [64]. This study focuses solely on predicting the degree of traffic congestion through information from a single source. Expanding the data set, with variables and resources from other sources such as speed, volume, times, meteorological factors, could improve the performance of the models [62].

2.3.5. GRU technique

Do et al., ITS stand out as an effective tool and measures to facilitate traffic management and solve traffic jam problems, among other uses in the transportation sector. They consider that an important point in the development of ITS systems is an appropriate prediction of the traffic flow, which in the last decade has had a laborious penetration. They point out that the traffic network has behaviors with dynamic spatial and temporal correlations, and some algorithms capture static dependencies. Taking into account that deep learning models based on attention have led to important advances in performance measures [35], they propose a traffic flow predictor based on deep learning with space–time attention. The attention model extracts the spatial characteristics between the regions of the road network structure and the temporal correlations of the time series. The proposed algorithm is statistically significant with a significance level of 99.99%, compared to other techniques such as SVR, GRU, LSTM and DCRNN, [43] The proposed model presents a mean RMSE prediction error of 9, 5%, MAE of 6.9% and WMAPE of 16.3%, lower than the compared reference models [35]. Although the aforementioned models have been shown to achieve high forecasting performance, one challenge to consider is getting the forecast results to achieve a compelling interpretation [42].

Li et al., note that recent deep learning models for traffic prediction do not take spatial structure into account and consider it a challenging task in terms of the changing dynamics of the road and its own difficulty in forecasting at long term. Therefore, they model a diffusion recurrent convolutional neural network (DCRNN), incorporating space–time dependencies of the vehicular flow, with this, the spatial characteristics are extracted through random bidirectional paths within the graph, and the temporal characteristics through timed sampling based on in a timed sampling codec [43]. When evaluating the proposed method in two sets of traffic data METR-LA and PeMS in California, a constant improvement of 12% to 15% is observed with respect to other reference algorithms, such as HA, ARIMA, VAR, SVR and LSTM. In the METR-LA data set, the proposed method obtains a MAPE of 7.3%, 8.8% and 10.5% in ranges of 15, 30 and 60 min, respectively. Likewise, in PeMS, the DCRNN method obtains a MAPE of 2.9%, 3.9% and 4.9% for the 15, 30 and 60 min intervals, respectively [43]. This research uses measures of static distance or road network topology for correlation between locations. Likewise, they omit the periodic effect

of the long-term incidence and the variation over time of the data, in the forecast [55].

Da Zhang et al., discusses relevant elements from external sources that could improve the performance of the traffic flow forecast, and that are generally ignored, such as climatic factors. Based on the above, Zhang et al., proposes a deep GRU network model (DGRNN), with hidden deep GRU layers [6]. In the model, elements of the hidden layer are substituted for GRU units to extract correlations between both long-term and short-term historical data, the authors say. A last layer groups the previous characteristics to make the forecast of the traffic flow. A total of 11 variables are analyzed from data obtained from the PeMS traffic information system and data on meteorological characteristics from the repository of the National Oceanic and Atmospheric Administration. It is observed that the precision results of the proposed algorithm exceed 94% for periods of two months and two years. The model results show an improvement in prediction precision of up to 5.8%, compared to the results of other models, with the same hyperparameters [6]. The authors also indicate that combining PeMS data with meteorological conditions favors forecast accuracy. However, the data set used to conclude the results is small; PeMS purvey hourly sampled data.

2.3.6. GCN technique

Recent research indicates that the structure of the road network can be formulated as a graph. The changing dynamics of the very nature of traffic flow and its correlation with an adaptive model for long-term forecasting could still be improved. Non-linear time frames as time moves into the future is still a challenge [65]. To do this, Zheng et al., proposes a Multiple Attention Graphics Network (GMAN) to forecast traffic conditions on a graph of the road structure. GMAN follows a codec architecture, where the first encodes the characteristics of the incoming traffic and the decoder performs the forecast of the data stream. The proposed model adds a layer of attention between the two previous elements to convert the coded features and create the future representations. The encoder and decoder, made up of attention blocks, model the dynamics of the adaptive spatio-temporal representations and are integrated through a closed fusion mechanism. The model, when evaluated against other recent techniques, presented the lowest prediction error, in the 60 min interval, with MAE, MAPE and RMSE values of 1.86, 4.31% and 4, 32, respectively, in the established PeMS data. During a 15 min interval, Graph WaveNet outperforms the proposed method. However, its results are encouraging. Although graph-based algorithms have achieved significant improvements over methods that do not use graph structure, they do not take advantage of the manifest spatial dependencies between variables [66].

For Zhao et al., they highlight traffic prediction as a challenging task due to its complex spatial and temporal dependencies, and indicate that the dynamic behavior of traffic flow, changing over time, is related to the physical structure of the vehicular network. CNN-based models, which have been shown to be effective over other methods and are commonly used for Euclidean data, have drawbacks for representing spatial correlations in traffic networks with complex structures [67]. Therefore, Zhao et al., propose a model based on graphs for the traffic forecast that allows extracting characteristics of the road network taking into account the spatial dependence. A closed recurring unit is introduced to obtain the dynamic variation of the traffic time series to model the time dependence. The proposed approach is evaluated on the Luohu district taxi speed data sets in Shenzhen and the Los-loop data set. The results show a reduction in the prediction error by approximately 1.5% to 57.8% compared to all the reference methods. The best precision result was 91.27% in Los-Loop for a 15 min interval. Likewise, the MAE and RMSE metrics were evaluated with results of 3.1802 and 5.1264 respectively, for the same interval and data set. MAE and RMSE in SZ-Taxi for a 15 min interval was 2.7061 and 3.9162 respectively. However, these investigations construct the graphs from the physical structure of the traffic network, without taking into account other factors that could improve the interpretation of the flow (eg, the type of road, the historical behavior, functionality of the region). Effective extraction of these interpretations could help enhance the forecast results [68]. Although the study correlates spatial dependencies between network nodes, they do not take into account other external parameters, such as meteorological factors, occurrence of latent events or incidents in traffic [69].

Zhu et al., contemplate that to improve forecasting effectiveness, the greatest amount of information from various sources that can be obtained from the traffic flow should be gathered, highlighting not only historical data, but a wide variety of external factors correlated. They highlight that recently, the spatio-temporal models that integrate GCN and RNN have become provisional models for traffic prediction. However, the integration of historical information with external factors is very limited [70]. For this reason, Zhu et al., propose a space–time GCN with increased attributes. The authors take into account the external conditions that affect mobility, and model their characteristics in relation to the traffic data at the nodes of the road network to obtain characteristic vectors with more information. In this way, they observe improvements in the perception of the external information model, which is reflected in the precision of the forecast [70]. The model is evaluated in the SZ Taxi dataset that contains taxi trajectory data in the city of Shenzhen, information of categorized points of interest, through SZ _POI, and climatological information through SZ _Weather. The evaluation results are compared with other models, including: SVR, ARIMA, GCN, GRU, DCRNN and TGCN. For an interval of 15 min., The precision of the proposed model was 71.93%. The precision error based on MAE and RMSE obtained was 2.7035 and 4.0294 is the interval of 15 min.

Xie et al., takes advantage of the impact of incidents that occurred on the road network to improve the prediction results of vehicle flow speed [16]. First, they design a high-impact critical traffic incident discovery method. Second, a deep learning model is designed to capture the characteristics of the incident. A GCN with Deep Incident Recognition (DIGC-Net) is established to effectively incorporate the characteristics of traffic incidents, space–time, periodic and context for the forecast of traffic speed. The experiment is run on traffic data sets drawn from the city of San Francisco and New York. When compared to different methods such as ARIMA, SVR, LSTM, ConvLSTM, STDN, and RNN, the proposed technique is the one that achieves the best forecasting results. The model has an MAPE of 11.02% in SFO and 17.21% in NYC. Likewise, there is a relatively improvement in the performance of periodic spatio-temporal learning, including incidents compared to data that did not take this aspect into account. The evaluation using the MAPE metric in the dataset that did not take into account the characteristics of incidents was 12.47% in SFO and 18.69% in NYC [71]. However, the model captures the dependency of temporal features and spatial features separately, which can limit prediction performance, since possible interactions between features [72] are ignored.

Bai et al., finds that current GCN techniques employ predefined graphs to extract spatial representations, and makes this perceptible to the quality of the graph. These graphs are often spontaneous and incomplete for forecasting studies, and cannot be adapted to domains without proper knowledge [73]. Therefore, Bai et al., propose to improve the GCN with two adaptive modules: the first to extract characteristics of the node. The second, to capture the spatial characteristics of the nodes and produce the respective graph that the training will execute. The experiment is evaluated on the PeMS data set. The prediction performance of the proposed model shows improvements in precision of more than 5%, compared to other methods such as HA, VAR, GRU, DCRNN, STGCN, ASTGCN and STSGCN. The best results, based on the prediction error metrics MAE, MAPE and RMSE were established at 15.95, 10.09% and 25.22 respectively. Furthermore, the training times, for a total of 150386 parameters, were 33.88 s [73].

For Chen et al., traffic forecasting still presents challenges in terms of complicated space–time dependence and inherent uncertainty caused by the very nature of the road's topological structure and various factors that influence mobility . Recent studies model the spatial features by GCN using a graph with fixed weights [74]. Chen et al., proposes a convolutional graphical model of attention with multiple range of two components. In the first instance, they establish the node and edge graph in relation to the distance of the structure from the network. Then the convolutional network captures the interactions of the nodes and edges. Finally, a multiple range care model learns the characteristics of the different neighborhood ranges. The experimental results, based on data extracted from the traffic information systems METR-LA and PeMS, show that the proposed model achieves promising results. The performance of the proposed model was compared with other reference models such as LSTM, Graph WaveNet and DCRNN, for forecasts of 15, 30 and 60 min in advance. Based on the MAE, RMSE, and MAPE metrics, the experiment yielded prediction errors of 2.67, 5.12, and 6.8% respectively, for a 15-minute interval, and 3.49, 7.30, 10.0% for a 60-minute interval [74]. According to [74], the proposed approach can be extended to model spatio-temporal correlations that may associate other conditions that affect traffic volume, such as points of interest and accidents. Furthermore, optimal neural architectures are considered to differ between different cities. Recent studies generally incorporate residual networks to add features in contiguous layers, without mixing features in the deep and shallow layers [39].

For Fu et al., the current models that use GCN omit the correlation between the structure of the network graph and the traffic flow information, considering the structure of the graph as a physical topology of the vehicular traffic network, which also omits the uncertainty [75]. They propose a convolutional network of Bayesian space–time graphs for the forecast of the traffic flow, introducing uncertainty to the traffic data for the construction of the graph structure, starting from a graph topology that learns both the physical structure and the data from end-to-end traffic detector nodes, to discover more accurate description of the relationship between traffic flows [75]. The performance of the proposed model is verified in two sets of real data, and the results show that the proposed algorithm achieves a better accuracy to recent models. The prediction results, based on the MAE and RMSE metrics are 2.6490 and 3.9670 respectively. The precision performance was 72.37% for a 15 min interval in the SZ Taxi data set. For this database, the AST-GCN method proves to have better precision performance than the proposed model. However, for the Los-Loop data set, the results are superior to the other reference methods, in all the horizon windows evaluated, with MAE, RMSE and Precision of 2.9150, 4.7585 and 91.85% in the interval of 15 min. Despite achieving impressive results, BSTGCN still cannot predict the value of the points that change sharply.

Recent studies have used various intelligent deep learning algorithms to forecast traffic in urban areas, exploiting spatial dependency using GCN techniques and extracting temporal characteristics using recurrent neural networks. However, Zhou et al. considers that the spatial characteristics are being represented statically and the periodicity of the flow volume during multiple instants of time has not been taken into account [76]. Zhou et al., foresees as an important challenge to take into account the dynamics of space–time variations and at the same time consider the influence of factors not related to traffic, such as the type of day of the week or the hour. Based on this, they propose a reinforced spatio-temporal attention model. This model uses a diffusion network in a graph to add neighboring information used to extract the dynamic patterns of the spatial correlations. Likewise, temporal dependencies are represented through a recurring neural network [76]. Model performance produces the best results, compared to other reference methods such as FC-LSTM, DCRNN, Graph WaveNet, Mres-RGNN, and DST-GCNN. The MAE, RMSE and MAPE evaluation metrics in the model yield results of 2.63, 5.09 and 6.88% for a 15 min time window, and 3.36, 7.29 and 9.97 % for a 60 min iteration, respectively, in METR-THE. Likewise, for the data set, the PeMS, MAE, RMSE, and MAPE prediction results are 1.22, 2.65, and 2.62% for a 15-minute window, and 1.71, 4.42, and 4.465 for a 60-minute interval. However, the lack of uncertainty and confidence in the modeling with respect to the expected results becomes the obstacle to prevent existing methods from using practical applications [77]. A key factor in traffic flow forecasting models is to consider external characteristics as relevant indicators. However, simply linking the above items end-to-end can cause feature overlap that could lead to unfortunate forecast results. Recognizing the importance of individual characteristics could facilitate effective data grouping and increase the performance of the model [78].

Zhang et al. consider traffic forecasting as an indispensable part of an ITS, and find challenging aspects in this task, such as: First, the traffic data is physically associated with road networks and therefore must be formatted as traffic graphs rather than as regular grid tensioners. Second, traffic data is highly spatially dependent, which implies that nodes in traffic graphs often have complex and dynamic relationships with each other. Third, the traffic data demonstrates a strong time dependence, which is crucial for modeling traffic time series [79]. Thus, they propose a convolutional model that extends the CNN network to graph domains and know the structure of graphs for traffic prediction. They propose two convolutional modules to capture the global and local structures respectively and are integrated to build an end-to-end network for traffic forecasting. Furthermore, in this process, pseudo three-dimensional (P3D) convolution networks are combined with SLCs to capture time dependencies in traffic data. Experiments were widely compared in six data sets where they show better results, compared to models such as DCRNN [43], GWaveNet [44] and STGCN [80]. For example, in the PeMS data set collected for six months using 325 sensors, it achieves a MAPE result of 3.0% for a time window of 15 min., And in the METR-LA data set collected for three months with 107 sensors, you get a MAPE of 6.7% for the same interval.

3. Literature review analysis

This section is divided as follows: Section 3.1 discusses the revised prediction techniques, in terms of evaluation metrics and data sets used. In Section 3.2 the opportunities and challenges still existing in the forecasting algorithms are detected. And finally, in Section 3.3 a scheme for the prediction of traffic flow is proposed, taking into account the opportunities detected.

3.1. Analysis

Table 1 summarizes the traffic flow prediction results obtained in previous studies, by various authors. The evaluation metrics most used by the authors are precision performance, Mean Absolute Percentage Error (MAPE), Mean Square Error (RMSE) and Mean Absolute Error (MAE). The results show the best precision of each of the methods, with short-term forecasts, with a horizon window of up to 20 min. According to the MAPE metric, the best prediction results of the traffic flow forecast techniques are obtained by ST-Traffic-Net, GMAN and DCRNN, however, according to the literature, these results depend on the database selected, its dimensionality and the same evaluation metric. We can see in the Table that the models that use deep learning and combine these techniques through hybrid algorithms achieve promising results in the prediction of traffic flow, superior to traditional parametric and machine learning techniques.

In recent years, various authors have explored different techniques and models for the prediction of vehicular flow, using from statistical models such as Kalman filters and ARIMA, to intelligent algorithms where the neural network is used in multiple ways. Lately, the development of deep learning algorithms has been explored in intelligent mobility applications for the prediction of traffic flow, showing substantial improvements compared to previous algorithms [43,50]; In addition, authors have explored hybrid models that combine two or

Table 1
Traffic flow prediction results from previous studies.

Year	Author	Technique	Dataset	Indicator	Evaluation metric				
					ACC (%)	MSE	MAE	MAPE (%)	RMSE
2016	Cai et al. [81]	KNN	Car speed in Liuliqiao District, Beijing	Speed				6.99	0.05
2016	Chen [58]	AutoEncoder	Traffic accident, Japan	Accidents			0.96		1.00
2016	Wu and Tan [59]	CNN-LSTM	PeMS	Volume			19.37	7.36	
2016	Fu et al. [82]	LSTM-GRU	PeMS	Volume		710.05	18.127		
2017	Fang, et al. [83]	DNN	HTC Corporation	Transport mode	95.70				
2017	Fouladgar et al. [25]	CNN-LSTM	PeMS	Speed	80				
2017	Jia et al. [18]	R-DBN, R-LSTM	BTMB and NMC, China	Volume			11.69		240.98 veh/h
2017	Du et al. [52]	CNN-LSTM	PeMS	Volume					0.028
		GRU						0.033	
		LSTM						0.033	
		RNN						0.04	
2017	Zhang et al. [37]	ST-ResNet	TaxiBJ and BikeNYC	Volume					6.33
2017	Ma et al. [84]	CNN	Beijing dataset (Taxi GPS)	Speed	94.20	22.825			
2018	Li et al. [43]	DCRNN	METR-LA & PeMS	Speed			1.38	2.90	2.95
		HA				2.88	6.80	5.59	
		FC-LSTM				2.05	4.80	4.19	
		ARIMA				1.62	3.50	3.3	
		VAR				1.74	3.60	3.16	
		SVR				1.85	3.80	3.59	
2018	Yu et al. [64]	STGCN	PeMS	Speed			1.36	2.90	2.96
2018	Wu, et al. [34]	CNN-RNN	PeMS	Speed			0.07	19.12	27.9183
2018	Cheng, et al. [26]	Adaptive-STKNN	PeMS	Speed			0.09	13.50	0.12
		STKNN				0.10	14.80	0.133	
2018	Zhao, et al. [67]	GCN [85]	SZ_taxi	Speed	72.43		2.59		5.06
		T-GCN			91.27		2.70		3.91
2018	Da Zhang et al. [6]	Deep GRU RNN (DGRNN)	PeMS and NOAA	Speed			0.0079		0.0019
2018	Duan et al. [86]	CNN-LSTM	Xián taxi GPS trajectory	Trajectory		11.1519			
2018	Yao et al. [38]	CNN - LSTM	Didi Service, Guangzhou, China	Taxi demand				16.16	9.642
2018	Wang et al. [45]	DBL	PeMS	Volume				3.87	124.31
		P-DBL					3.23	110.43	
2018	Wu et al. [34]	CNN-RNN (DNN-BTF)	PeMS	Volume			19.13		27.92
2019	Tang et al. [29]	SVM+EEMD	TDRL, Minnesota	Volume			8.03	6.26	10.63
2019	Do et al. [35]	STANN	VicRoads	Volume			6.5	15.60	8.9
2019	Wang et al. [47]	LSTM	Automatic vehicle identification, Xuancheng, China	Speed		13.92			
2019	Liu et al [33]	Gradient boosting decision tree	LPR in Nanjing, China	Volume				9.69	
2019	Lin et al. [36]	Conv+	TaxiBJ - BikeNYC	Flow trajectory			22.94		336.29
2019	Tempelmeieret al. [19]	SVR	Data4 Urban Mobility	Speed			0.469		1.024
2019	Wei et al. [57]	AutoEncoder	PeMS	Volume			16.15		26.32
2019	Hou et al. [17]	ARIMA-WNN	Wenhuadong intersection/Tongyi, Weihai City, China	Volume				5.12	5.72
2019	Impedovo et al. [40]	WaveNet Convolution	PeMS y TRAP-2017	Volume			6.229	15.52	8.668
		LSTM				6.274	15.70	8.734	
		GRU				6.323	17.76	8.637	
		SAE				6.095	16.74	8.335	
2019	Emami et al. [14]	Faded Memory Kalman Filter	AIMES, Australia	Volume					0.18
2019	Tao et al. [24]	Delay-based Spatial–Temporal ARMA	England Highway traffic flow data	Volume		1017.9		11.49	
2019	Yao et al. [55]	CNN - LSTM	TaxiNYC, BikeNYC	Flow trajectory			16.3	24.1	
2019	Yu et al. [80]	LSVR	METR-LA & PeMS	Speed			1.42	3.31	3.45
		ST-Unet				1.38	2.79	2.83	
2019	Chen et al. [87]	Residual Recurrent Graph	PeMS	Speed			1.52	3.2	3.23
			METR-LA			2.07	4.9	3.57	
2019	Guo et al. [88]	ASTGCN	PeMS	Flow			16.63		25.27
2019	Wu et al. [44]	Graph WaveNet	METR-LA and PeMS	Speed			1.3	2.73	2.74
2019	Zhang et al. [62]	Autoencoder	Google Map,Bing map, Autonavi map, Beijing Traffic	Congestion		0.0579	0.0106		
2020	Ata, et al. [5]	SVM	Congestion		98.70				
2020	Alotaibi [89]	Ensemble Learning	TDM, Poland	Transport mode	95.60				
2020	Liu et al. [31]	Chaotic particle swarm SVM	Guizhou, China	Volume		1.8481			

(*continued on next page*)

Table 1 (*continued*).

Year	Author	Technique	Dataset	Indicator	Evaluation metric				
					ACC (%)	MSE	MAE	MAPE (%)	RMSE
2020	Siddiqui [61]	Deep Extreme Learning Machine	PKLot, Curitiba, Brazil	Parking space	94.37				
2020	Li et al. [39]	ST-NASNet	TaxiBJ	Flow				25.97	15.88
2020	Sun et al. [15]	CNN-BiLSTM	Urumqi traffic network	Speed				15.0	19.97
2020	Wang et al. [51]	LSTM	Guiyang, Guizhou province, China	Volume					23.115
2020	Vijayalakshmi et al. [53]	CNN - LSTM	PeMS	Volume			2.3		3.3
2020	Shen et al. [20]	RVR-CSA	TDRL, Minnesota	Volume			1.06–23.45	4 a 15	5.98–29.62
2020	Lin et al. [21]	RVM	Whitemud Drive, Canadá.	Volume			41.09	13.83	6.41
2020	Bratsas [22]	SVR	Traffic Thessaloniki (Grecia) and Floating Car	Speed			6.25		
2020	Kouziokas [50]	LSTM y Bi-LSTM	PeMS	Volume					6.53
		LSTM						9.31	
		Bi-LSTM						6.53	
		SVM						12.39	
		ANN						15.46	
2020	Ai et al. [90]	Fuzzy-RBF	Beijing Transportation Research Center	Congestion			1.89		2.15
2020	Xie et al. [71]	DIGC-Net	HERE Taxi Traffic API	Incident				11.02	
2020	Lu et al. [42]	ST-TrafficNet	PeMS & METR_LA	Speed			1.26 veh	2.68	2.72
2020	Zheng et al. [65]	GMAN	Xiamen and PeMS	Traffic condition			1.34	2.81	2.82
2020	Zhu et al. [70]	AST-GCN	SZ taxi, POI, and Weather	Speed	71.93		2.7035		4.0294
2020	Bai et al. [73]	AGCRN	PeMS			15.95	Volume	10.09	25.22
2020	Song et al. [91]	STSGCN	PeMS	Volume			17.13	10.96	26.86
2020	Chen et al. [74]	MRA-BGCN	PeMS & METR-LA	Speed			1.29	2.90	2.72
2020	Fu et al. [75]	BSTGCN	SZ-Taxi & Los-Loop	Speed			2.649		3.967
2020	Zhou et al. [76]	RSTAG	PeMS & METR-LA	Speed			1.22	2.62	2.65
2020	Zhang [79]	SLC-CNN	PeMS	Volume			2.22	5.21	4.07
			PeMS-BAY			1.44	3.0	2.90	
			METR-LA			2.53	6.7	5.18	
2021	Majumdar et al. [12]	LSTM	Traffic data for Buxton, UK	Speed	94.6				

more techniques, where the benefits of each are extracted to enhance the model and increase the accuracy of the forecast.

Figs. 2, 3 and 4 show the most outstanding results of the algorithms used for the traffic flow forecast, taking into account the three most used data sets in the review: PeMS, METR-LA and SZ-Taxi, to have a better approximation of your results.

Data-driven methods such as ARIMA and Kalman filtering are still popular, however they are generally based on the assumption of stationarity, which is often violated by traffic data. Taking into account the non-linear and stochastic features of the urban traffic flow condition, algorithms based on artificial neural networks and machine learning better exploit the above difficulties. Figs. 2, 3 and 4 show that the accuracy of these statistical models is lower than those based on machine learning.

Several authors agree that the key point in the forecast of traffic flow is in the treatment of spatio-temporal characteristics. Because the sequence of spatial characteristics of the historical data defines the behavior of the time series data, it clearly affects the development of the next trend. That is, nearby nodes tend to have a hard connection to each other [52].

In addition to the obvious spatial and temporal features of the traffic flow, there are several factors that affect traffic dynamics and congestion, such as traffic accidents, road conditions, weather conditions, events, points of interest, among others. As the data source becomes more heterogeneous, deep learning algorithms are projected as relevant alternatives, for its ability to represent the multilayered perceptual structure similar to the transmission of information between the neurons of the human brain to recognize nonlinear data patterns.

The evaluation metrics most used by the authors to evaluate the performance of the traffic flow prediction techniques in urban areas are the Mean Square Error (MSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Square root of MSE, precision (ACC) and confusion matrix.

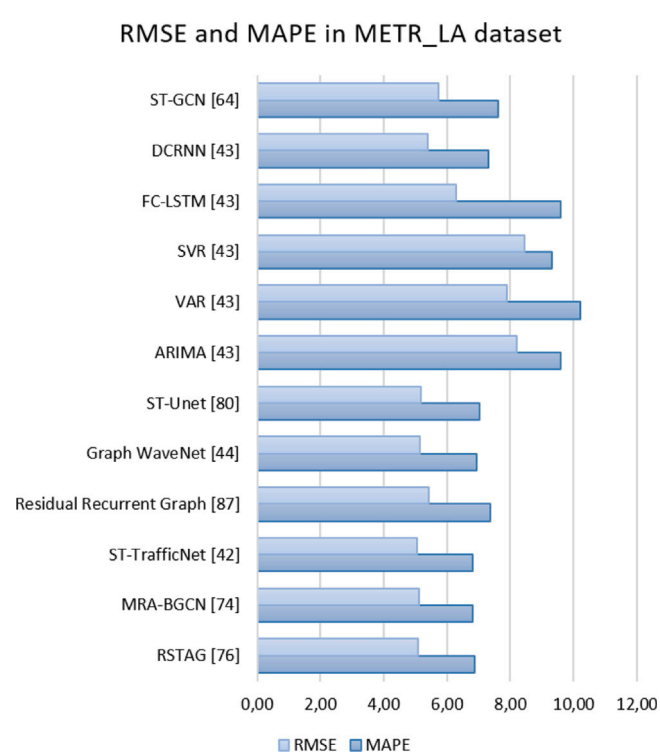

Fig. 2. Results of traffic prediction by MAPE and RMSE metrics, from studies using METR_LA dataset, for traffic speed indicator.

MAE is a metric that equals to the mean absolute value of the difference between the true values y_i and the predicted value $\hat{y}_i$. Its

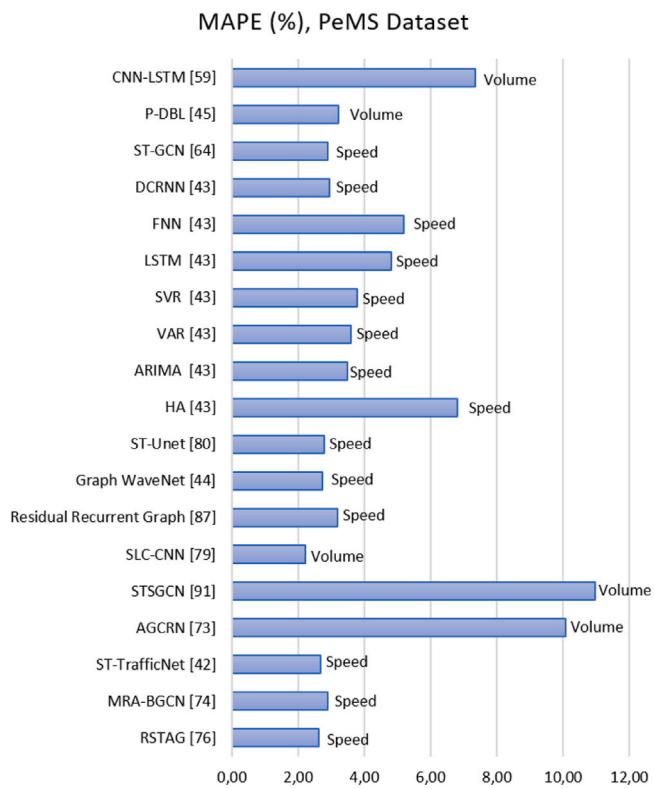

Fig. 3. Results of traffic prediction by MAPE metrics, from studies using PeMS dataset, for traffic volume and speed indicators.

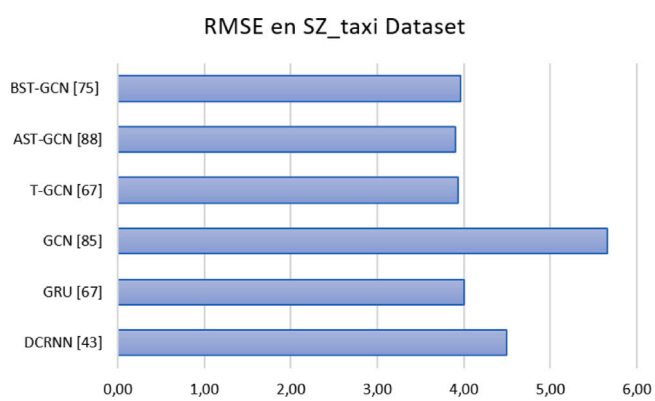

Fig. 4. Results of traffic prediction by MAPE metrics, from studies using SZ-Taxi database, for traffic speed indicator.

definition is given as:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{1}$$

MAPE is a performance indicator that measures the size of the error in percentage terms. It is defined as follows:

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i - \hat{y}_i|}{y_i} * 100\% \tag{2}$$

MSE is one of the simplest metrics for prediction evaluation and calculates the squared difference between predictions and observations, then averages their values. It is defined as follows:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{3}$$

Table 2
Confusion matrix.

	True	False
True	TP	FP
False	FN	TN

RMSE is the standard deviation of the forecast error presented by the concentration of the data around the prediction result. It is defined as follows:

$$RMSE = \frac{1}{N}\sqrt{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{4}$$

where y_i are the observed values and $\hat{y}_i$ are the predicted values for the interval i, and N is the size of the sample used in the evaluation.

ACC is the most common metric for summarizing overall ranking precision for all classes and is defined as:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

Where TP indicates the true positives, given by the number of positive cases that were classified as correct; TN denotes the True Negatives, given by the number of negative cases that were classified as negative; FP denotes the false positives, given by the number of negative cases that were classified as positive; and FN denotes the false negatives, given by the number of positive cases that were classified as negative.

Confusion Matrix is a 2×2 table that reports true positives, false positives, true negatives, and false negatives. This table allows to visualize in detail the performance of the classification algorithm, describing how the actual values and predictions are distributed. The Table 2 shows its representation.

3.2. Opportunity area detected

With the benefits of capturing and representing spatio-temporal characteristics in data sets that use time series, intelligent algorithms based on deep learning have obtained better prediction performance in vehicular flow data. However, although the temporal and spatial correlations have been taken into account in these algorithms, there are still challenges to be faced:

- The spatial dependency can be dynamic in time [55]. The spatial dependency between locations is based on the historical similarity of the traffic [37,55] and the algorithm learns a static spatial dependency. However, the dynamics between various nodes or areas can vary over time.
- Traffic data is not strictly periodic [55], and may vary according to other factors such as weather conditions, traffic accidents or scheduled events.
- For the spatio-temporal prediction, the city is generally divided into a grid map that describes a region, and the spatial characteristics that model the range of neighborhood dependencies are focused on local information; convolutions model spatial dependency in different ranges, with empirical and fixed-size kernel [39]. The lack of clarity of these models for the assignment of weights to the neural units in the hidden layers, as well as the number of input nodes, prevents us from explaining the relationship between the inputs and the results of the forecast of the historical data flow. [92].
- They have difficulties to function under the scheme of a road network structure with complex topology, so, in effect, they have difficulties to describe the spatial dependence.
- The investigations carried out in this field construct the graphs from the road structure, and do not take into account other aspects that could be used to semantically correlate the traffic

network, such as the type of road, the functional environment of a specific area or its historical behavior, which would help improve the effectiveness of the forecast [68].

The correlations of the nature of the traffic flow are dynamic over time. To address the complicated spatial dependencies of road networks, since it is naturally structured as a graph in a non-Euclidean space with roads as nodes and their natural connections as edges, recent research is employing the road structure as a graph. The GCNs, which have performed efficiently within the traffic flow modeling addressing spatio-temporal graphs and handling of non-Euclidean correlations, are integrated with the RNN [43] or the CNN [80] to incorporate historical knowledge of the vehicular network and extract the characteristics that correlate the paired nodes. Recent studies have addressed diverse deep learning techniques for traffic prediction, for example, taking advantage of GCN to characterize spatial dependence and using RNN to capture temporal dynamics.

Although neural networks that use graphs to represent space–time dependencies have shown significant precision results compared to others that do not use graph structure, as shown in Figs. 2, 3 and 4 there are still challenges to face:

- The neural networks based on graphs to represent the space–time dependencies determine the structure of the graph based on the physical topology of the road map, disregarding the possible correlation with the traffic data [75]; therefore, it is not guaranteed to be an optimal description of the dependency between traffic flows.
- These models use predefined interconnection graphs to extract spatial dependencies, with measures of similarity or distance [54]. Likewise, These graphs are often spontaneous and incomplete for forecasting studies, and cannot be adapted to domains without proper knowledge [73].
- The introduction of uncertainty in the structure of the graph, such as the random drop of nodes or edges, could improve the generalizability of GCNs; the graph structure used in existing GCN-based methods is deterministic, meaning a lack of investigation of uncertainty [75].
- Some studies based on GCN, moreover, mainly focus on modeling spatial dependence using GCN on a graph with fixed weight values, however, the correlations between vertices by pairs, change with the current traffic condition [74].

3.3. COVID-19 in traffic flow prediction

At the end of 2019, the COVID19 pandemic emerged and most nations began to activate the mechanisms to prevent the spread of the virus in an accelerated manner. The declaration of a state of emergency and the intervention of local politics evidenced a significant reduction in vehicle flow density in many regions of the world. Studies carried out in cities such as New York, Rome, New Delhi, Wuhan and Tokyo [93], the Czech Republic [94], Greece [95], the city of Reggio Emilia, Italy [96] and Poland [97] coincide in the reduction of vehicular flow, due to confinement and compliance with social distancing recommendations. Likewise, the way of getting around changed during the emergency measures, with the use of public transport being reduced to give way to the use of individual transport, mainly the private car [98].

The unexpected changes that COVID-19 has caused in daily life have added unforeseen dependencies, which influence vehicular mobility in cities. Liapis, [99] conducted tests with vehicle flow data before and during the lockdown. The results show that the lockdown and post-lockdown periods turned out to be the most difficult to predict (76% and 75%), compared to normal prediction values on pre-pandemic days (84%), using random forest algorithms and logistic regression.

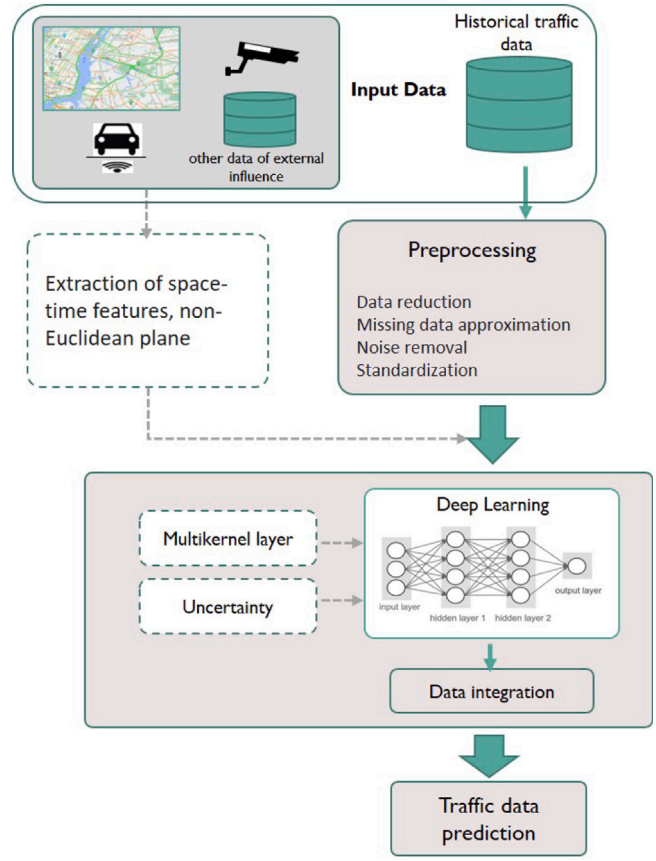

Fig. 5. Traffic flow prediction model scheme.

Lv, [100], evaluates algorithms such as GCN, LSTM,GRU, AST-GCN [80], T-GCN [88] and DCRNN [43] with urban traffic revitalization index data from 29 Chinese cities, provided by the Didi Platform, and highlight the relevance of linking COVID19 epidemic data that includes suspected, positive people with the disease, cured people, and deaths, both in the domain of space and time. Taking this information into account, the algorithm proposed by Lv demonstrates better prediction performance, with MAPE differences greater than 2% compared to those previously mentioned.

We have seen how the effects that the pandemic has left us have had a notable impact on the way we move, from political decisions to restrict mobility to the perception of security under the fear of epidemic contagion to choose a means of transport, also considering the information population epidemic. Aspects that could be added and taken into account in the development of traffic flow prediction algorithms to try to approximate the model to the current mobility conditions to increase the levels of precision that improve decision making.

3.4. Structure of a traffic flow prediction system

The Fig. 5 illustrates a general scheme for the prediction of vehicular traffic flow, which consists essentially of three stages: data collection, pre-processing and a neural network structure for the processing and predictive analysis of traffic flow, mainly deep, Given the remarkable results, stop at other techniques.

In the data collection we find the input data that are considered to forecast the traffic condition, within which there is spatio-temporal information on the current and historical traffic flow of the urban area of study, provided by networks of sensors, detectors on the road, information sent by GPS of the vehicles, among others. Data sources of

external influence such as weather information, road conditions, event scheduling in influence zones, points of interest, accident information, can help the prediction results of the algorithm, adding in these times of pandemic, epidemic information and partial or total mobility restriction measures.

The collected and transformed data is pre-processed to refine unstructured, imbalanced, or non-standardized data points; noise is removed to help reduce network traffic and latency, and normalized to scale the data according to the standard normal distribution.

The processing scheme analyzes the information using data analysis algorithms powered by machine learning models to extract the required key information. Deep learning algorithms can extract spatial and temporal dependencies more efficiently. A multiple kernel function can assign different weights to neural units and could help to model the range of spatial dependency between neighboring regions that may be different. Likewise, an algorithm that allows inferring the uncertainty incorporated by external traffic flow data could increase the performance of the model.

Then, through a unified data integration service, information links are generated, in such a way that the merged data sets, from multiple sources, are prepared to be consulted in search of information.

The model takes into account the main challenges found in the literature and can help improve accuracy performance, taking advantage of the proven ability of deep learning algorithms in handling both spatial and temporal dependencies of the vehicular network state. Likewise, by enhancing the model through dynamic weighting coefficients, it is possible to learn local and global characteristics of the nodes of the topological structure that help to exploit the space–time dependencies of the traffic more closely. Finally, considering heterogeneous data from multiple sources, including palpable information evidenced by the unexpected changes that the pandemic has left us, we can obtain new space-temporal dependencies not considered to date, which affect mobility and could improve the accuracy rates of the forecasting algorithm.

4. Conclusions

In this document we carry out a review of intelligent techniques for the predictive analysis of traffic flow in urban areas. Techniques recently used in the literature were identified and categorized. We not only investigated the precision results of the different models, but also briefly described the methodology used to understand the benefits and limitations of the applied techniques. The results of quantifiable precision of the different models were compared to obtain an overview of their effectiveness, highlighting their benefits and limitations to generalize the challenges still present in the area of urban traffic prediction. From there we can say that both models based on deep learning and hybrid algorithms achieve promising results for the prediction of urban traffic, superior to traditional models such as ARIMA, SVR and intelligent KNN and SVM algorithms. Among them, GCN has achieved remarkable results for its ability to address the complicated spatial dependencies of road networks, given their natural graphical structure. Recent efforts have taken advantage of GCN to extract spatial characteristics and RNN to extract temporal dynamics, however, there are still challenges that could be addressed to improve prediction results, taking into account that aspects in the capture and extraction of dependencies can be addressed spatial–temporal external data that affect mobility conditions, such as weather, events, points of interest and social networks. Likewise, consider that the range of spatial dependence may be different between spatially close or neighboring regions, whose characteristics vary with the traffic condition, and can help to provide dynamic information about the urban traffic network.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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